

Assignment 1
Operating System (BCAC0022)
BCA (III Sem.) 2nd Year

1. Define operating system and list the basic services provided by operating system.
2. Explain essential features of following structure of O.S
 - a) Monolithic System
 - b) Micro Kernels
3. Explain the process state transition diagram used in multiprogramming environment. Describe the fields in a process control block (PCB). What is switching overhead?
4. Write short notes on the following: (a) Time Sharing System; (b) Real-Time System.
5. What is thread? Explain classical thread model OR Explain threads in detail
6. List the main difference and similarities between threads and process
7. Consider following processes with length of CPU burst time in milliseconds

Process	Burst time
P1	5
P2	10
P3	2
P4	1

All process arrived in order p1, p2, p3, p4 all time zero

- a) Draw Gantt charts illustrating execution of these processes for SJF and round robin (quantum=1)
 - b) Calculate waiting time for each process for each scheduling algorithm
 - c) Calculate average waiting time for each scheduling algorithm.
8. Explain the following term related to IPC: a) Race condition b) critical region
9. What are critical sections? Why mutual exclusion required? Explain any 2 methods of achieving mutual exclusion in detail
10. Define semaphore and discuss product-consumer problem and Dining philosopher problem with semaphore.
11. Explain Process Control Block (PCB)
12. Define thread. Explain thread structure.
13. Five batch jobs A to E arrive at same time. They have estimated running times 10,6,2,4 and 8 minutes. Their priorities are 3,5,2,1 and 4 respectively with 5 being highest

priority. For each of the following algorithm determine mean process turnaround time. Ignore process swapping overhead. Round Robin, Priority Scheduling, FCFS, SJF.

14. Assume arrival order is: P1, P2, P3, P4, P5 at time 0, 1, 2, 3, 4 respectively and a smaller priority number implies a higher priority. Draw the Gantt charts for preemptive and non-preemptive priority scheduling. Calculate Average Turnaround Time and Average Waiting Time.
15. Define semaphore and Give the implementation of Bounded Buffer Producer Consumer Problem using Semaphore
16. Define Deadlock and list the conditions that lead to deadlock. How Deadlock can be prevented